Weather-Driven Late Blight Risk Model: Project Report

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Summary

For this project I wanted to see whether free weather data could tell me when potatoes in Norfolk were at risk of late blight. I downloaded 21 seasons of hourly weather (2005 to 2025), coded the two warning rules UK growers actually use, and checked the results against years when blight was reported in the farming press.

The short answer is that the rules worked reasonably well, but rainfall did slightly better. Season rainfall lined up with all 6 reported years, the Hutton Criteria with 5 and the older Smith Period with 4. Both rules missed 2012, which was a very bad blight year nationally. I also set up a machine learning step, but I couldn't train it because I didn't find any dated disease records.

Biology of late blight

Before writing any code I read up on the disease itself, because the warning rules only make sense once you know what the pathogen needs. Late blight is caused by *Phytophthora infestans*. It looks like a fungus but is actually an oomycete, or water mould, and it is still the most expensive potato disease in Great Britain.

It spreads in a cycle that can repeat many times in one season:

1. **Sporulation.** Infected leaves produce spores (sporangia) when humidity stays around 90% or higher for several hours, usually overnight.
2. **Dispersal.** Wind and rain splash carry the spores onto new leaves and into nearby fields.
3. **Infection.** The spore needs a film of water on the leaf. In cooler weather (roughly 10 to 15 °C) it releases swimming zoospores; in warmer weather (20 to 25 °C) it germinates directly.
4. **New lesions.** Symptoms appear 3 to 7 days later, and those lesions start producing spores of their own.

So the key conditions are mild nights, high humidity and wet leaves. The pathogen slows down below about 10 °C and during hot, dry spells. Because every cycle adds more spores, several good days in a row matter much more than one.

Data and methods

I picked one site in Norfolk (52.45° N, 1.30° E) because it is one of the main potato-growing areas in England. The weather came from NASA POWER, which is free and gives hourly values for any coordinates. I needed hourly rather than daily data because both rules count hours of high humidity. In total that came to 77,112 hourly records covering May to September of each season.

- I used three variables: air temperature, relative humidity (both at 2 m) and rainfall.
- I turned each day into a few numbers: minimum, maximum and mean temperature, mean humidity, the number of hours at 90% humidity or above, and total rainfall. I dropped any day with fewer than 20 hours of data.
- A rule counts as met on a day when both that day and the previous day qualify.

Rule	Min temperature (each day)	Hours at RH ≥ 90% (each day)	Notes
Smith Period (1956)	10 °C or more	11 or more	The old UK standard
Hutton Criteria (2017)	10 °C or more	6 or more	What AHDB uses today
Near miss	9 °C or more	5 or more	Days that just miss Hutton, similar to BlightWatch's amber warning

Everything was done in Python (pandas, NumPy, matplotlib and scikit-learn) in a Jupyter notebook. The code is on my GitHub as *late-blight-risk-model*.

Features and their biological basis

I wanted every input to have a biological reason behind it, so that I could explain the model rather than just report a score.

Feature	What it stands for	Why it matters
Minimum temperature of 10 °C or more	Spore production and infection	Nights need to be warm enough for the pathogen to grow
Hours at 90% humidity or more	Spore production and infection	The closest thing I had to leaf wetness
Two qualifying days in a row	Repeated cycles	One humid night isn't enough to start an epidemic
Rainfall over the last 7 days	Spread	Rain splashes spores around and keeps leaves wet
Favourable days in the last 7	Build-up	Each cycle adds more spores
Average max temperature over 7 days	Slowing down	Hot spells hold the pathogen back
Day of the year	Crop stage	Later in the season the canopy closes and traps humidity

The first three are the published rules. I built the rolling features for the machine learning part.

Results

The first thing I noticed was how often the Hutton Criteria fired. On average it flagged 63 days per season out of 153, compared with 16 for the Smith Period. That is around 4 in every 10 days, which is far more warnings than a real service would send out.

- Hutton days ranged from 35 (2015) up to 99 (2021).
- Smith days ranged from just 2 (2009) to 37 (2021). Because it needs 11 humid hours instead of 6, it separates good and bad years much more clearly.
- The first Hutton warning each year fell somewhere between 2 May and 14 June, which is roughly when growers would start spraying.
- On the risk calendar, May is mostly quiet, risk picks up in June, and July to September is where it really clusters.

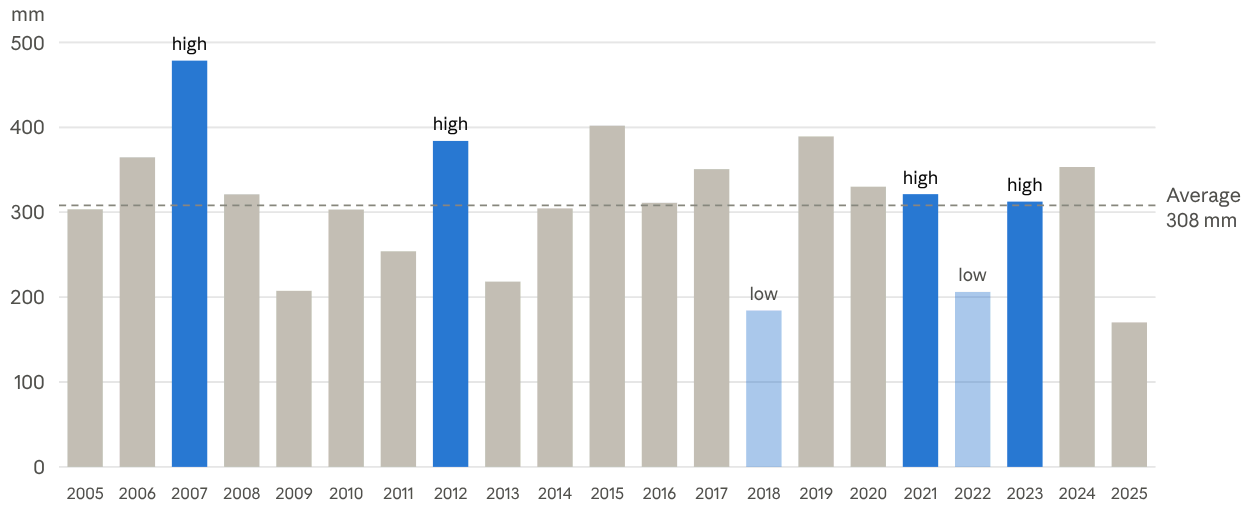
I think the high Hutton count says more about the data than about the disease. The grid cell sits close to the coast, where the air stays humid for long stretches.

Validation against reported seasons

To check whether any of this matched reality, I searched farming news and advisory reports for seasons where blight pressure was clearly described as high or low, and found six. I then compared each one with the 21-season average: a high year should sit above it and a low year below it. Rainfall got all 6 right, Hutton got 5 and Smith got 4.

Season rainfall matched all 6 reported blight years

May–September rainfall per season, Norfolk. Labelled bars: blight pressure reported in published sources.



NASA POWER hourly data, summed in the project notebook · 21 seasons, 2005–2025

Season	Reported pressure	Evidence	Hutton days (avg 63)	Smith days (avg 16)
2023	High	120 outbreaks by mid-August after the wettest July since 2009 (Farm Contractor)	83	14
2022	Low	37 outbreaks mapped by late August in a dry summer (Farmers Guardian)	47	13
2021	High	Weather favoured blight, nights often above 10 °C; Scotland-focused (Agriland)	99	37
2018	Low	No cases in Scotland and very few UK-wide, credited to dry weather (FAS)	55	12

Season	Reported pressure	Evidence	Hutton days (avg 63)	Smith days (avg 16)
2012	High	335 outbreaks by mid-August against 183 in 2011; "extreme disease pressure" (Farmers Weekly)	54	11
2007	High	Record 301 confirmed outbreaks, more than double any earlier season (PotatoPro)	68	29

Both rules missed 2012, which surprised me because it was one of the worst blight years on record. Farmers Weekly reported that by 11 July 2012 there had been more than five times as many Smith Periods nationally as in 2011, but my Norfolk point only showed 11 against 8. One grid point clearly wasn't picking up what was happening across the country.

I should also be honest that rainfall's 6 out of 6 is not as strong as it sounds. In 2021 and 2023, rainfall was only 4 to 13 mm above the average, so a slightly different cut-off would have flipped those two years.

What could and could not be validated

I could only check the results at the level of whole seasons. The other two checks I had planned both needed dated disease records, which I couldn't find.

Check	Done?	What happened
Rules coded as published	Yes	Thresholds match Smith (1956) and Dancey (2018), and I tested the logic on hand-made examples
Season-level comparison	Yes, 6 seasons	Hutton 5/6, Smith 4/6, rainfall 6/6
Did a warning come 7 to 21 days before an outbreak?	No	No dated outbreak records found

Check	Done?	What happened
Machine learning on real disease data	No	The pipeline is ready (logistic regression and random forest, split by year) but has nothing to learn from yet

Things to keep in mind:

- NASA POWER is gridded data with cells about 50 km wide, not a weather station in a field. Humidity inside a crop canopy can be quite different, and one point can't represent a whole region.
- The rules only tell you whether the weather suits the disease. Whether an outbreak actually happens also depends on spores being present, the potato variety and fungicide use.
- Six seasons is a small sample, so none of this is statistically strong.

Conclusion and next steps

Overall, I was able to rebuild the official UK blight rules from free data and get results that mostly line up with what growers reported. The most interesting finding for me was that rainfall seemed to add something the humidity-based rules miss. That fits the biology, since rain is what spreads the spores. With only six seasons, though, I would call it a promising lead rather than a proven result.

If I take this project further, this is what I would do next:

1. Compare NASA POWER with a Met Office weather station near the site, to see how far off the humidity is.
2. Run the model for an inland site and a few other regions, to see whether 2012 shows up.
3. Add a rainfall condition to the Hutton rule and test it again.
4. Ask AHDB's Fight Against Blight service for dated outbreak records, so I can test the warnings properly and finally train the machine learning model.

References

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- Blight update shows positive signs, Farmers Weekly, 15 August 2012.
- Crop Health Updates – July 2018, Farm Advisory Service.
- High disease pressure for cereals and potatoes across the UK, Agriland, 8 July 2021.
- Blight monitoring shows 6_A1 strain prevalent this season, Farmers Guardian, 26 August 2022.
- UPL advises growers on potato blight following wettest July in over a decade, Farm Contractor, 21 August 2023.
- Code and data: github.com/junaid18afzal-code/late-blight-risk-model